

M.S. Ramaiah Institute of Technology
(Autonomous Institute, Affiliated to VTU)
Department of Artificial Intelligence and Data Science

Course Name: Introduction to Artificial Intelligence(AI)

Course Code: AI35/AD35

Credits: 3:0:0

UNIT -3

Introduction

Automated Planning –

- Definition of Classical Planning ([Chapter 11: 11.1](#))
- Uncertainty: Acting under Uncertainty ([Chapter 12: 12.1](#))
- Basic Probability Notations: What probabilities are about ([Chapter 12: 12.2](#))
- Bayes' Rule and its Use ([Chapter 12: 12.5](#))
- Applying Bayes' rule: The simple case
- Naive Bayes Models ([Chapter 12: 12.2](#))
- Text classification with naive Bayes([Chapter 12: 12.6](#))

Introduction

- **Intelligent robots** or **self-driving cars** or **smart cities**, they will all use different aspects of Artificial Intelligence!!!
- But to create such AI model, Planning is very important.
- Planning is required for every task(Job)

For example, **reaching a particular destination** requires planning.

- Finding the best route and then identify a set of actions to be done at a particular time is very important.

Definition

Planning researchers have invested in a **factored representation** using a family of languages called **PDDL**, the Planning Domain Definition Language. **Types of PDDL**

1. Classical planning is a basic PDDL to define the task of finding a sequence of **actions** to accomplish a goal in a discrete, deterministic, static, fully observable environment.

2. Extensions can handle non-classical domains that are continuous, partially observable, concurrent, and multi-agent.

- **Planning Domain Definition Language(PDDL)** a language used to represent all actions into one action schema.

Definition

- In PDDL, a **state** is represented as a conjunction of ground atomic and fluents.
- “**ground atomic**” means there is a single predicate, and if there are any arguments, they must be constants.
- “**fluent**” means an aspect of the world that changes over time

Definition

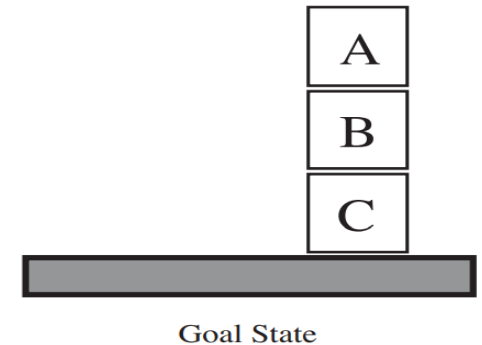
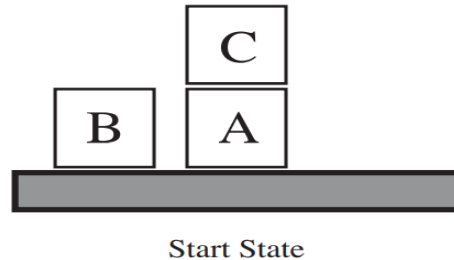
PDDL describes the four basic things needed in a search problem:

- Initial state
- Actions
- Result
- Goal

The Blocks world Problem

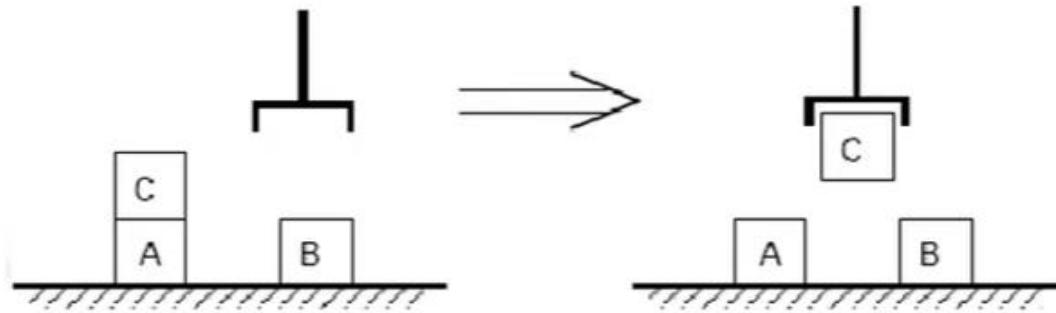
PDLL describes the four basic things needed in a problem

- Initial state
- Actions
- Result
- Goal: To build one or more stacks of blocks
block A on B and block B on C



The Blocks world Problem

The Blocks world consists of:



A flat surface such as a table top

An adequate set of identical blocks which are identified by letters.

The blocks can be stacked one upon another.

There is a robot arm that can manipulate the blocks.

The robot can hold one block at a time and only one block can be moved at a time.

Any number of blocks can be on the table.

The Blocks world Problem

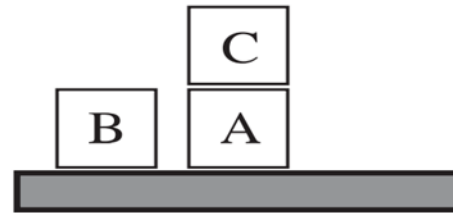
- To represent the blocks world using PDDL, we need to define states and actions.
- To define states and actions, we need to specify constants and predicates.
- What are our constants? A, B, C, Table.
- What are our predicates?

The Blocks world Problem

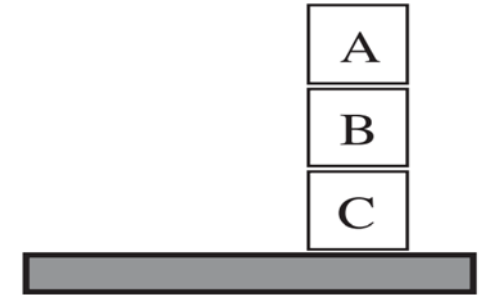
Actions

- **Move(b, x, y)**
 - Move block b from the top of x to the top of y
 - No other block should be on it
- **MoveToTable(b, x)**
 - Move a block b from x to the table
 - No other block should be on it
- **Clear(x)**

There is a clear space on x to hold a block

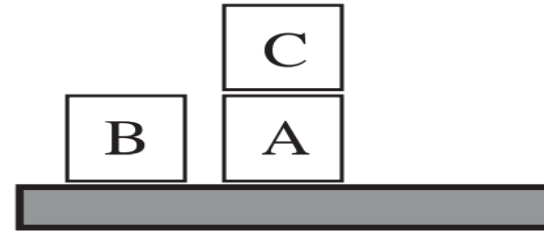


Start State

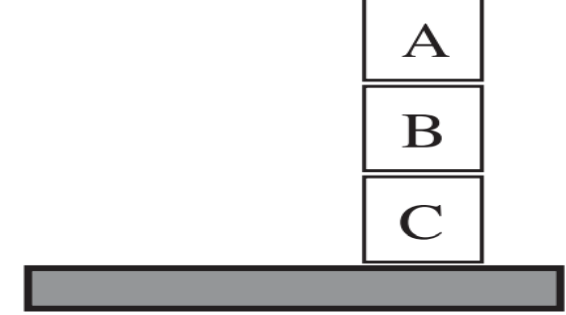


Goal State

The Blocks world Prc



Start State



Goal State

Build a three-block tower

One solution is the sequence

[

MoveToTable(C, A),

Move(B, Table, C),

Move(A, Table, B)

]

$$\begin{aligned}
 &Init(On(A, Table) \wedge On(B, Table) \wedge On(C, A) \\
 &\quad \wedge Block(A) \wedge Block(B) \wedge Block(C) \wedge Clear(B) \wedge Clear(C)) \\
 &Goal(On(A, B) \wedge On(B, C)) \\
 &Action(Move(b, x, y), \\
 &\quad PRECOND: On(b, x) \wedge Clear(b) \wedge Clear(y) \wedge Block(b) \wedge Block(y) \\
 &\quad \quad (b \neq x) \wedge (b \neq y) \wedge (x \neq y), \\
 &\quad EFFECT: On(b, y) \wedge Clear(x) \wedge \neg On(b, x) \wedge \neg Clear(y)) \\
 &Action(MoveToTable(b, x), \\
 &\quad PRECOND: On(b, x) \wedge Clear(b) \wedge Block(b) \wedge (b \neq x), \\
 &\quad EFFECT: On(b, Table) \wedge Clear(x) \wedge \neg On(b, x))
 \end{aligned}$$

Air cargo transport

Air cargo transport problem

- Loading and unloading cargo
- Flying it from one place to other place.

The following plan is a solution to the problem

Air cargo transport

Actions:

- **Load:** This action is taken to load cargo.
- **Unload:** This action is taken to unload the cargo when it reaches its destination.
- **Fly:** This action is taken to fly from one place to another.

Therefore, the Air cargo transport problem is based on loading and unloading the cargo and flying it from one place to another.

Air cargo transport

The actions affect two predicates:

$\text{In}(c, p) \rightarrow$ cargo c is inside plane p

$\text{At}(x, a) \rightarrow$ object x (either plane or cargo) is at airport a

At really means “available for use at a given location.”

$[\text{Load}(C_1, P_1, \text{SFO}), \text{Fly}(P_1, \text{SFO}, \text{JFK}), \text{Unload}(C_1, P_1, \text{JFK}),$

$\text{Load}(C_2, P_2, \text{JFK}), \text{Fly}(P_2, \text{JFK}, \text{SFO}), \text{Unload}(C_2, P_2, \text{SFO})]$.

Air cargo transport

Init($At(C_1, SFO) \wedge At(C_2, JFK) \wedge At(P_1, SFO) \wedge At(P_2, JFK)$
 $\wedge Cargo(C_1) \wedge Cargo(C_2) \wedge Plane(P_1) \wedge Plane(P_2)$
 $\wedge Airport(JFK) \wedge Airport(SFO)$)
Goal($At(C_1, JFK) \wedge At(C_2, SFO)$)
Action(*Load*(c, p, a),
 PRECOND: $At(c, a) \wedge At(p, a) \wedge Cargo(c) \wedge Plane(p) \wedge Airport(a)$
 EFFECT: $\neg At(c, a) \wedge In(c, p)$)
Action(*Unload*(c, p, a),
 PRECOND: $In(c, p) \wedge At(p, a) \wedge Cargo(c) \wedge Plane(p) \wedge Airport(a)$
 EFFECT: $At(c, a) \wedge \neg In(c, p)$)
Action(*Fly*($p, from, to$),
 PRECOND: $At(p, from) \wedge Plane(p) \wedge Airport(from) \wedge Airport(to)$
 EFFECT: $\neg At(p, from) \wedge At(p, to)$)

Figure 10.1 A PDDL description of an air cargo transportation planning problem.

Definition

$$\text{Init}(\text{On}(A, \text{Table}) \wedge \text{On}(B, \text{Table}) \wedge \text{On}(C, A) \\ \wedge \text{Block}(A) \wedge \text{Block}(B) \wedge \text{Block}(C) \wedge \text{Clear}(B) \wedge \text{Clear}(C))$$

Initial state

It is the representation of **each state as the conjunction** of the ground and functionless atoms.

“Ground” means that no variables are used. $\text{On}(B, \text{Table})$ is legal

Example : $\text{At}(x, y)$ is illegal, because it uses variables x, y .

“Functionless” means that no functions are used.

Example: $\text{At}(\text{Father}(\text{George}), \text{JFK})$ is illegal, because it uses function Father .

Definition

Actions

It is defined by a **set of action schemas** which implicitly define the **ACTION()** and **RESULT()** functions.

- Each **action** has its own set of preconditions
- All the **preconditions** should be satisfied before performing the action
- Some **effects** can be positive or negative

Action schema:

- ACTION: specifies name and parameter list
- PRECONDITION: conjunction of positive literals
- EFFECT: conjunction of literals (positive or negative)

Action(Fly (p, from, to) ,

PRECOND: At (p, from) $\wedge$ Plane (p) $\wedge$ Airport (from) $\wedge$ Airport (to)

EFFECT: $\neg$ At (p, from) $\wedge$ At (p, to))

Action(Fly (P1, SFO, JFK) ,

PRECOND:At (P1, SFO) $\wedge$ Plane (P1) $\wedge$ Airport (SFO) $\wedge$ Airport (JFK)

EFFECT: $\neg$ At (P1, SFO) $\wedge$ At (P1, JFK))

Definition

Result

It is obtained by the set of actions used by the agent.

For example, with the action $\text{Fly}(P1, SFO, JFK)$, we would remove the fluent $\text{At}(P1, SFO)$ and add the fluent $\text{At}(P1, JFK)$.

Goal

Is a **conjunction of literals** (whose value is either positive or negative).

For example, the goal $\text{At}(C1, SFO) \wedge \neg \text{At}(C2, SFO) \wedge \text{At}(p, SFO)$, refers to any state in which cargo C1 is at *SFO* but C2 is not, and in which there is a plane at *SFO*

Definition

There are examples which will make PDDL understandable:

1. Have cake and eat cake too
2. The Blocks world
3. Air cargo transport
4. The spare tire problem

Formulate the PDDL description of any planning problem.

Have Cake and eat Cake too

The “have cake and eat cake too” problem

Init(Have(Cake))

Goal(Have(Cake) $\wedge$ Eaten(Cake))

Action(Eat(Cake))

PRECOND: *Have(Cake)*

EFFECT: $\neg$ *Have(Cake)* $\wedge$ *Eaten(Cake)*

)

Action(Bake(Cake))

PRECOND: $\neg$ *Have(Cake)*

EFFECT: *Have(Cake)*

)

The spare tire problem

Consider the problem of changing a flat tire

Initial state → Flat tire on the axle and Good spare tire in the trunk.

Goal → Good spare tire properly mounted onto the car's axle

Assume that the car is parked in a particularly bad neighborhood, so that the effect of leaving it overnight is that the tires disappear.

Actions: 4

1. removing the spare from the trunk
2. removing the flat tire from the axle
3. putting the spare on the axle
4. leaving the car unattended overnight

Init(*At*(*Flat*, *Axle*) $\wedge$ *At*(*Spare*, *Trunk*))

Goal(*At*(*Spare*, *Axle*))

Action(*Remove*(*Spare*, *Trunk*),

PRECOND: *At*(*Spare*, *Trunk*)

EFFECT: $\neg$ *At*(*Spare*, *Trunk*) $\wedge$ *At*(*Spare*, *Ground*))

Action(*Remove*(*Flat*, *Axle*),

PRECOND: *At*(*Flat*, *Axle*)

EFFECT: $\neg$ *At*(*Flat*, *Axle*) $\wedge$ *At*(*Flat*, *Ground*))

Action(*PutOn*(*Spare*, *Axle*),

PRECOND: *At*(*Spare*, *Ground*) $\wedge$ $\neg$ *At*(*Flat*, *Axle*)

EFFECT: $\neg$ *At*(*Spare*, *Ground*) $\wedge$ *At*(*Spare*, *Axle*))

Action(*LeaveOvernight*,

PRECOND:

EFFECT: $\neg$ *At*(*Spare*, *Ground*) $\wedge$ $\neg$ *At*(*Spare*, *Axle*) $\wedge$ $\neg$ *At*(*Spare*, *Trunk*)
 $\wedge$ $\neg$ *At*(*Flat*, *Ground*) $\wedge$ $\neg$ *At*(*Flat*, *Axle*))

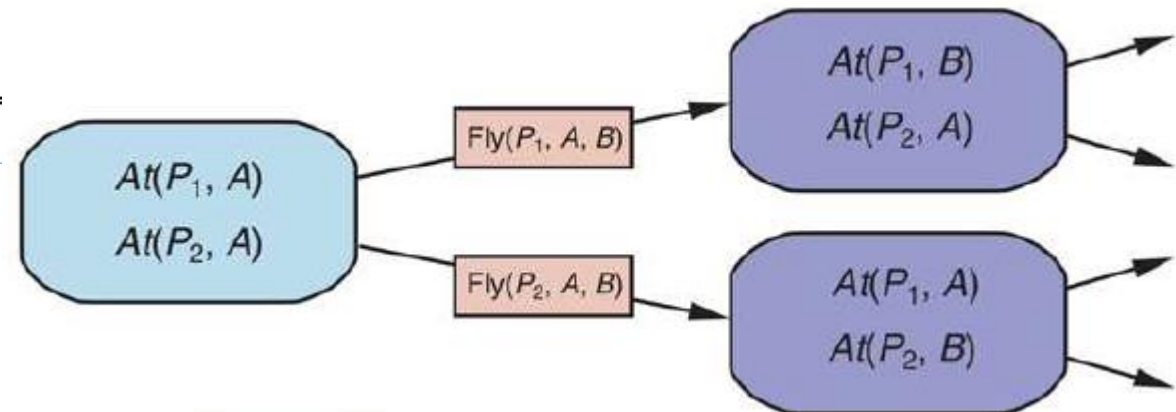
Algorithms for Planning as State-Space Search

Forward state-space search (Progression)- Air cargo problem

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Init( $At(C_1, SFO) \wedge At(C_2, JFK) \wedge At(P_1, SFO) \wedge At(P_2, JFK)$ 
 $\wedge Cargo(C_1) \wedge Cargo(C_2) \wedge Plane(P_1) \wedge Plane(P_2)$ 
 $\wedge Airport(JFK) \wedge Airport(SFO)$ )
Goal( $At(C_1, JFK) \wedge At(C_2, SFO)$ )
Action( $Load(c, p, a)$ ,
  PRECOND:  $At(c, a) \wedge At(p, a) \wedge Cargo(c) \wedge Plane(p) \wedge Airport(a)$ 
  EFFECT:  $\neg At(c, a) \wedge In(c, p)$ )
Action( $Unload(c, p, a)$ ,
  PRECOND:  $In(c, p) \wedge At(p, a) \wedge Cargo(c) \wedge Plane(p) \wedge Airport(a)$ 
  EFFECT:  $At(c, a) \wedge \neg In(c, p)$ )
Action( $Fly(p, from, to)$ ,
  PRECOND:  $At(p, from) \wedge Plane(p) \wedge Airport(from) \wedge Airport(to)$ 
  EFFECT:  $\neg At(p, from) \wedge At(p, to)$ )
  
```

Figure 10.1 A PDDL description of an air cargo transportation planning problem.



Forward state-space search (Progression)- Air cargo problem

- Forward State Space Planning (FSSP) is a subset of classical planning.
- In this approach, the planning system starts from the initial state and explores all possible actions to progress toward the goal.
- It systematically explores the state space by applying actions and transitions from one state to another until the goal is achieved.

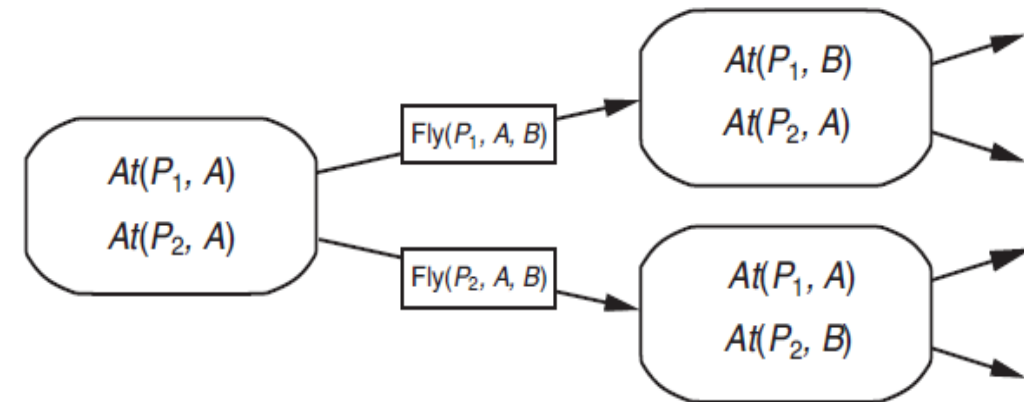
Key Features:

Exploration Direction: Starts from the initial state and moves forward toward the goal.

Guarantees: The algorithm is sound and guarantees a solution if one exists.

Challenges: FSSP can suffer from a large branching factor, meaning the number of potential actions can grow exponentially, leading to high computational costs.

Example: Solving a maze by exploring all possible paths from the starting point.



Backward state-space search (Regression)

Backward State Space Planning (BSSP) is another variant of classical planning, but it works in reverse.

It starts from the goal state and works backward to find the sequence of actions that leads to the initial state.

This approach is often more efficient in certain cases where the goal state is clearly defined and specific actions need to be reversed to reach the initial state.

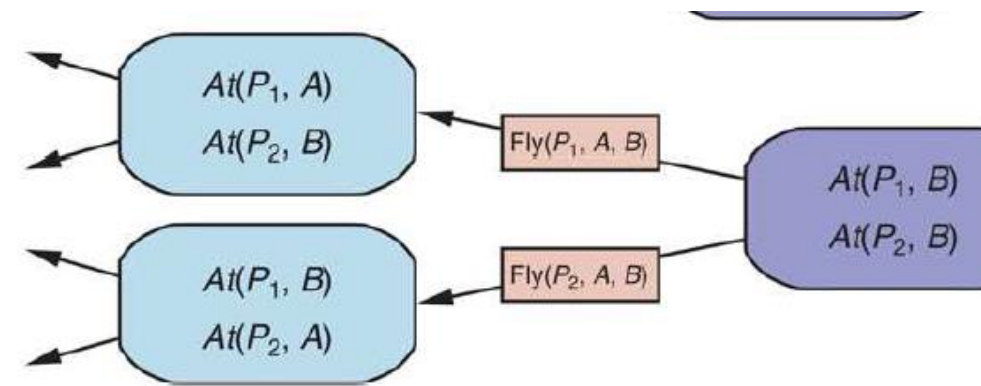
Key Features:

Exploration Direction: Starts from the goal and works backward toward the initial state.

Efficiency: BSSP often has a smaller branching factor, making it more computationally efficient.

Challenges: The algorithm may not always be sound, and inconsistencies might arise, leading to failure in finding a solution.

Example: Planning the steps needed to achieve a goal in chess, working backward from the checkmate position.



Acting Under Uncertainty

Agents in the real world need to handle **uncertainty**, as agent may never know for sure what state it is in now or where it will end up after a sequence of actions.

- The different causes of uncertainty are,
 1. The environment is **not fully observable**.
 2. The environment behaves in a **non-deterministic** way.
 3. The **actions** performed may **not have desired effect**.
 4. There might be **unjustified reliance on assumptions**.



Acting Under Uncertainty

Problem-solving and logical agents handle uncertainty by keeping track of a **belief state**—a representation of the set of **all possible world states** that it might be in—and generating a plan that handles every possible eventuality that its **sensors may report during** execution.

This approach works on simple problems, but it has **drawbacks**:

1. The agent must consider ***every possible explanation*** for its sensor observations, no matter how unlikely. This leads to a **large belief-state full of unlikely possibilities**.
2. A **correct plan** that handles every eventuality can grow arbitrarily large and must consider arbitrarily unlikely contingencies.
3. Sometimes there is **no plan that is guaranteed to achieve the goal—yet** the agent must act. It must have some way to compare the merits of plans that are not guaranteed.

Why do we need to consider uncertainty?

Agents never have access to the whole truth about the environment. Hence they must act under uncertainty.

Example for Uncertainty in AI

Automated taxi has the goal of delivering a passenger to the airport on time.

Plan **A₉₀** The agent forms a plan, that involves leaving home 90 minutes before the flight departs and driving at a reasonable speed.

Even though the airport is only about 5 miles away, a logical taxi agent will not be able to conclude with certainty that the plan will definitely work.

Uncertainties: Instead, it reaches the weaker conclusion that plan will work as long as the car doesn't break down or run out of gas, and I don't get into an accident, and there are no accidents on the bridge, and the plane doesn't leave early, and no meteorite hits the car, and . . .
..”

Thus, None of the actions can be deduced for certain

Consider the problem as follows

Let action A_t = leave for airport t minutes before flight .

Will A_t get me there on time?

Problems to consider :

- partial observability (road state, other drivers' plans, etc.)
- noisy sensors (traffic reports)
- uncertainty in action outcomes (flat tire, etc.)
- immense complexity of modeling and predicting traffic

Solution

Performance Measures :

- Getting to the airport in time.
- Avoiding long, unproductive wait at the airport.
- Avoid speeding tickets along the way (Must drive below the speed limit).

Based on this assume that A_{90} is expected to maximize the agent's performance measure. The agent's knowledge cannot guarantee any of these outcomes for A_{90} , but it can provide some degree of belief that they will be achieved.

Solution

Other Plan: such as A_{180} , might increase the agent's belief that it will get to the airport on time, but also increase the likelihood of a long, boring wait.

The right thing to do—the rational decision—therefore depends on both the relative importance of various goals and the likelihood, and degree to which, they will be achieved.

Example for Uncertainty in AI

Diagnosis—whether for medicine, automobile repair, or whatever—almost always involves uncertainty.

Consider an example of uncertain reasoning:

Diagnosing a dental patient's toothache.

Example for Uncertainty in AI

We consider a dental diagnosis using FOL.

Let us now consider the rule ,

$$\textit{Toothache} \Rightarrow \textit{Cavity}.$$

This is wrong as Not all patients with toothaches have cavities; some of them have gum disease, an abscess, or one of several other problems:

$$\textit{Toothache} \Rightarrow \textit{Cavity} \vee \textit{GumProblem} \vee \textit{Abscess} \dots$$

Example for Uncertainty in AI

In order to make the rule true, we have to add an almost unlimited list of possible problems. We could try turning the rule into a causal rule:

Cavity $\Rightarrow$ Toothache.

- But this rule is not right either; not all cavities cause pain.
- The only way to fix the rule is to make it logically exhaustive: to augment the left-hand side with all the qualifications required for a cavity to cause a toothache.

Example for Uncertainty in AI

- The theory of probability provides a way of summarizing the uncertainty that comes from our laziness and ignorance, thereby solving the qualification problem.
- We believe that there is, say, an 80% chance—that is, a probability of 0.8—that the patient who has a toothache has a cavity.
- This belief could be derived from statistical data—80% of the toothache patients seen so far have had cavities—or from some general dental knowledge, or from a combination of evidence sources.

Basic Probability Notation

- **Logical assertions** say which possible worlds are accepted and others strictly ruled out (all those in which the assertion is false).
- **Probabilistic assertions** talk about how probable the various worlds are.
- In probability theory, the set of all possible worlds is called the sample space.
- The possible worlds are mutually exclusive and exhaustive—two possible worlds both cannot be the case, and one possible world must be the case.

Basic Probability Notation

Random variable

- Variables in Probability theory are called Random Variables.
- A Random Variable is a variable that can take on different values randomly.
- Every random variable has a domain—the set of possible values it can take on.
- Variables can have infinite domains too either discrete (like the integers) or continuous (like the reals).
- Their names begin with an uppercase letter

Example :

D1 is random variable.

Domain of D1 is $\{1, \dots, 6\}$

Basic Probability Notation

1. Boolean Random Variables

Eg., Cavity (do I have a cavity?) It can take values : <true,false>

2. Discrete Random Variables

Eg., Weather is one of <sunny,rainy,cloudy,snow>

3. Continuous Random Variables

Eg., Height of different people in a given population set

Components of basic probability notation

1. *Unconditional probability*
2. *Prior probability*
3. *Evidence*
4. *Conditional probability*
5. *Posterior probability*

Basic Probability Notation

1. UnConditional Probability

- Unconditional probability, refers to a probability that is unaffected by previous or future events.
- Unconditional probability is simply the probability of an event occurring.
- Unconditional probability refer to degrees of belief in propositions in the **absence of any other information.**
- **Unconditional Probability of $A=P(A)$**

Basic Probability Notation

Examples of unconditional probabilities

- What is the probability of a dice rolling a 6?
 $A = \text{Dice rolling a 6}$ $P(A) = ?$
- What is the probability of landing a head on a coin toss? $B = \text{Landing a head on a coin toss}$
 $P(B) = ?$
- “What is the probability of rain tomorrow?”

Basic Probability Notation

2. Prior probability

- It shows the likelihood of an outcome in a given dataset.
- A priori probability refers to the likelihood of an event occurring when there is a finite amount of outcomes and each is equally likely to occur.
- **Priori Probability = Desired Outcome(s)/The Total Number of Outcomes**

Example:

Coin toss.

What is the a priori probability of a head in a single coin toss? $=1/2=50\%$.

Basic Probability Notation

Fair Dice Roll :A six-sided fair dice is rolled.

What is the a priori probability of rolling a 2, 4, or 6, in a dice roll?

The number of desired outcomes is 3 (rolling a 2, 4, or 6), and there are 6 outcomes in total. The a priori probability for this example is calculated as :

A priori probability = $3 / 6 = 50\%$. Therefore, the a priori probability of rolling 2, 4, or 6 is **50%**.

Basic Probability Notation

3. Conditional Probability

- A conditional probability, is the probability of an event which would affect or be affected by another event.
- A conditional probability, as the name implies, comes with a condition.
- Measure of the probability of an event **given that (by assumption, presumption, assertion)** another event has occurred.
- For example,
 - “What is the probability of rain tomorrow?”
A conditional probability can be phrased as follows:
“What is the probability of rain tomorrow, given today is sunny?”
 - What is the probability of rolling a 6 followed by a 4?

Basic Probability Notation

Conditional Probability: The probability of an event A , given that another event B has already occurred (i.e. A conditional B).

This is represented by $P(A|B)$ and we can define it as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability of A given B (red text) = $\frac{\text{Probability of } A \text{ and } B \text{ (blue text)}}{\text{Probability of } B \text{ (blue text)}}$

Basic Probability Notation

Example:

Susan took two tests. The probability of her passing both tests is 0.6. The probability of her passing the first test is 0.8. What is the probability of her passing the second test given that she has passed the first test?

Solution:

$$P(\text{second} | \text{first}) = \frac{P(\text{first and second})}{P(\text{first})} = \frac{0.6}{0.8} = 0.75$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Probability of A given B Probability of A and B Probability of B

Basic Probability Notation

Weather forecasters use conditional probability to predict the likelihood of future weather conditions, given current conditions.

For example, suppose the following two probabilities are known:

$$P(\text{cloudy}) = 0.25$$

$$P(\text{rainy} \cap \text{cloudy}) = 0.15$$

$$P(\text{rain} | \text{cloudy}) = P(\text{rainy} \cap \text{cloudy}) / P(\text{cloudy})$$

$$P(\text{rain} | \text{cloudy}) = 0.15 / 0.25$$

$$P(\text{rain} | \text{cloudy}) = 0.6$$

The probability that it will rain *given* that it is cloudy out, is 0.6 or **60%**.

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Probability of A given B

Probability of A and B

Probability of B

Basic Probability Notation

Sports betting companies to determine the odds they should set for certain teams to win certain games. Suppose the Probability of Team A star player is hurt is 0.15 and probability of Team A wins and Team A star player is hurt is 0.02. What is the probability that Team A will win *given* that their star player is hurt.

For example, suppose the following two probabilities are known about some basketball team:

$$P(\text{Team A star player is hurt}) = 0.15$$

$$P(\text{Team A wins} \cap \text{Team A star player is hurt}) = 0.02$$

The company could use these values to calculate the probability that team A will win, given that their star player is hurt:

$$P(\text{Team A Wins} | \text{star is hurt}) = P(\text{Team A Wins} \cap \text{star is hurt}) / P(\text{star is hurt})$$

$$P(\text{Team A Wins} | \text{star is hurt}) = 0.02 / 0.15$$

$$P(\text{Team A Wins} | \text{star is hurt}) = 0.13$$

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Probability of A given B

Probability of A and B

Probability of B

The probability that Team A will win *given* that their star player is hurt is 0.13 or **13%**.

Problem Statement: In a school, 40% of students take mathematics ($P(Math) = 0.4$), and 25% of the students who take mathematics pass the exam ($P(Pass|Math) = 0.25$). What is the probability a randomly selected student takes mathematics and passes the exam?

Solution: The probability a student takes mathematics and passes the exam is:

$$P(Math \cap Pass) = P(Math) \cdot P(Pass|Math)$$

Substitute the values:

$$P(Math \cap Pass) = 0.4 \cdot 0.25 = 0.1$$

Answer: The probability a randomly selected student takes mathematics and passes the exam is 10%.

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Probability of A given B

Probability of A and B

Probability of B

Problem Statement: An online store promises delivery within 2 days. 60% of orders are shipped on time ($P(OnTime) = 0.6$). If shipped on time, 90% are delivered within 2 days ($P(Delivered|OnTime) = 0.9$). If not shipped on time, only 20% are delivered within 2 days ($P(Delivered|NotOnTime) = 0.2$). What is the probability an order was shipped on time given it was delivered within 2 days ($P(OnTime|Delivered)$)?

Solution:

$$P(OnTime|Delivered) = \frac{P(Delivered|OnTime) \cdot P(OnTime)}{P(Delivered)}$$

where:

$$P(Delivered) = P(Delivered|OnTime) \cdot P(OnTime) + P(Delivered|NotOnTime) \cdot P(NotOnTime)$$

Substitute the values:

$$P(Delivered) = (0.9 \cdot 0.6) + (0.2 \cdot 0.4) = 0.54 + 0.08 = 0.62$$

$$P(OnTime|Delivered) = \frac{0.9 \cdot 0.6}{0.62} = \frac{0.54}{0.62} \approx 0.871$$

Answer: The probability the order was shipped on time given it was delivered within 2 days is about **87.1%**.



$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability of A given B

Probability of A and B

Probability of B

Basic Probability Notation

For any propositions a and b , we have

$$P(a | b) = \frac{P(a \wedge b)}{P(b)};$$

Conditional probability can be written in a different form called the Product rule:

$$P(a \wedge b) = P(a | b)P(b)$$

Basic Probability Notation

Product rule:

$$P(a \wedge b) = P(a | b)P(b)$$

Examples:

- If $P(A)=.5$ and $P(B|A)=.4$, $P(B \wedge A)=.4 \times .5 =.2$.
- $P(\text{King}|\text{Diamond}) = 1/13$, $P(\text{Diamond}) = 1/4$, therefore $P(\text{King} \wedge \text{Diamond}) = 1/13 \times 1/4 = 1/52$

Basic Probability Notation

- If the probability that a driver encounters a green light at the first intersection is 0.7 and the probability of encountering a green light at the second intersection is 0.6, and these events are independent, then what is the probability that the driver encounters green lights at both intersections.

$$P(G_1 \cap G_2) = P(G_1) \times P(G_2) = 0.7 \times 0.6 = 0.42$$

- If a customer places two separate orders online, the probability of the first order arriving on time is 0.9, and the probability of the second order arriving on time 0.85. Assuming these events are independent, what is the probability that both orders arrive on time is:

$$P(O_1 \cap O_2) = P(O_1) \times P(O_2) = 0.9 \times 0.85 = 0.765$$

- If two diagnostic tests are conducted independently, the probability that the first test detects a disease is 0.8, and the probability that the second test detects the disease is 0.7. Then what is the probability that both tests detect the disease. $P(D_1 \cap D_2) = P(D_1) \times P(D_2) = 0.8 \times 0.7 = 0.56$

Bayes' rule

Consider that A and B are any two events from a sample space S

Using our understanding of conditional probability, we have:

$$P(A|B) = P(A \cap B) / P(B)$$

$$P(B|A) = P(A \cap B) / P(A)$$

It follows that $P(A \cap B) = P(A|B) * P(B) = P(B|A) * P(A)$

Thus, $P(A|B) = P(B|A) * P(A) / P(B)$

This equation is known as Bayes' rule (also Bayes' law or Bayes' theorem). This equation underlies most modern AI systems for probabilistic inference.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Probability of A and B

Probability of A given B

Probability of B

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Likelihood

Prior

Posterior

Evidence

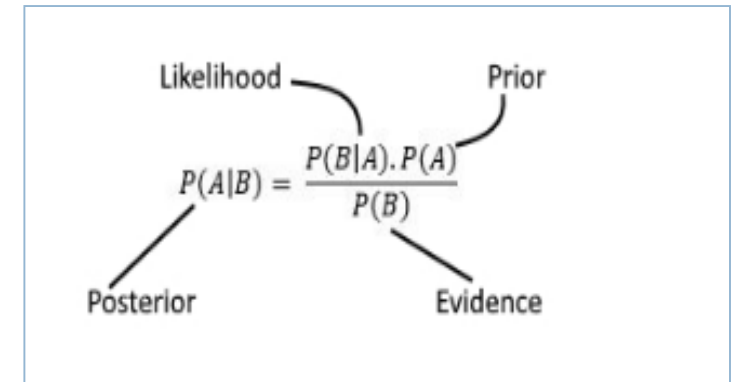
Bayes' rule

$P(A|B)$: The probability of event A (the hypothesis) given that B (the evidence) has occurred. This is the **posterior probability**.

$P(B|A)$: The probability of event B given that A is true. This is the **likelihood**.

$P(A)$: The probability of event A being true, without any new evidence. This is the **prior probability**.

$P(B)$: The probability of event B occurring, irrespective of A . This is the **Evidence**.



The diagram shows the formula for Bayes' Theorem: $P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$. Labels with arrows point to the components: 'Likelihood' points to $P(B|A)$, 'Prior' points to $P(A)$, 'Posterior' points to $P(A|B)$, and 'Evidence' points to $P(B)$.

Bayes' Theorem thus gives the probability of an event based on new information that is or may be related to that event. The formula also can be used to determine how the probability of an event occurring may be affected by hypothetical new information, supposing the new information will turn out to be true.

Basic Probability Notation

4. Posterior Probability/Bayes Rule:

- The posterior probability is the probability of event A occurring given that event B has occurred.
- The conditional probability that is assigned after the relevant evidence or background is taken into account.
- **Posterior = Prior * Likelihood**

The following is a more formal definition.

Definition Let A and B be two events whose probabilities $P(A)$ and $P(B)$ are known. If also the conditional probability $P(B|A)$ is known, Bayes' rule gives

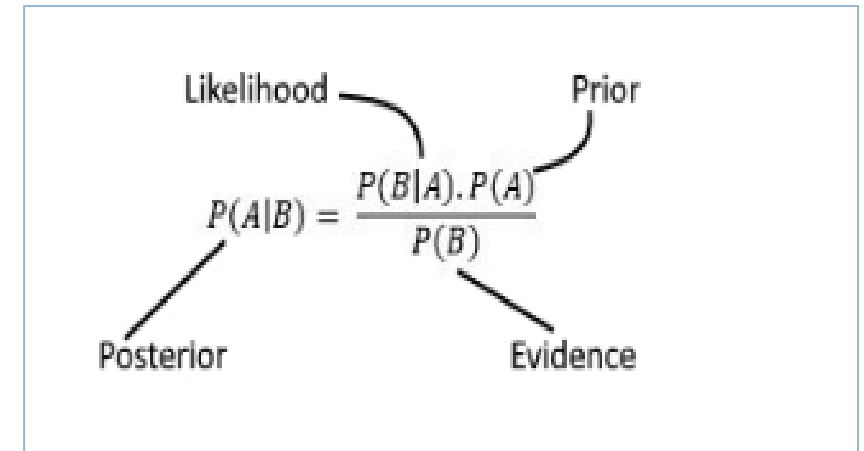
$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

The conditional probability $P(A|B)$ thus computed is called posterior probability.

Bayes' rule

Consider a group of people with 50% of them are employed ($P(E)=.5$) and 40% Women ($P(W)=.4$). Given that she is employed, the conditional probability of the selected individual being a woman is 20% ($P(W|E) = .2$). What is the probability that a person selected from the group is employed, given the person selected is a woman ($P(E|W) = ?$).

$$\begin{aligned} P(E|W) &= P(W|E) P(E)|P(W) \\ &= 0.20 \cdot .5 / 0.4 \\ &= 0.25 \\ &= 25\% \end{aligned}$$



Bayes' rule

From a standard deck of playing cards, a single card is drawn. The probability that the card is king is $4/52$, then calculate posterior probability $P(\text{King}|\text{Face})$, which means the drawn face card is a king card.

Solution:

Bayes theorem in Artificial intelligence

$$P(\text{king} | \text{face}) = \frac{P(\text{Face}|\text{king}) \cdot P(\text{King})}{P(\text{Face})}$$

Bayes' rule

P(king): probability that the card is King= $4/52 = 1/13$

P(face): probability that a card is a face card= $3/13$

P(Face|King): probability of face card when we assume it is a king = 1

Putting all values in equation we will get:

$$P(\text{king}|\text{face}) = \frac{1 * (\frac{1}{13})}{(\frac{3}{13})} = 1/3, \text{ it is a probability that a face card is a king card.}$$

Applying Bayes' rule: The simple case

- **Example:** A doctor knows that the disease meningitis causes a patient to have a stiff neck, say, 70% of the time. The doctor also knows some unconditional facts: the prior probability that any patient has meningitis is 1/50,000 and the prior probability that any patient has a stiff neck is 1%.
- **Solution:**
- Letting s be the proposition that the patient has a stiff neck and m be the proposition that the patient has meningitis, we have

$$P(s | m) = 0.7$$

$$P(m) = 1/50000$$

$$P(s) = 0.01$$

$$P(m | s) = \frac{P(s | m)P(m)}{P(s)} = \frac{0.7 \times 1/50000}{0.01} = 0.0014.$$

$$P(\text{cause} | \text{effect}) = \frac{P(\text{effect} | \text{cause})P(\text{cause})}{P(\text{effect})}.$$

A spam filter detects the word “discount” in emails. 70% of all emails are spam. Among spam emails, 40% contain the word "discount". Among non-spam emails, 10% contain the word "discount". If an email contains the word “discount,” what is the probability it is spam?

$$P(\text{Spam}) = 70\% = 0.7$$

$$P(\text{Discount}|\text{Spam}) = 40\% = 0.4$$

$$P(\text{Discount}|\text{NotSpam}) = 10\% = 0.1$$

$$P(\text{Spam}|\text{Discount}) ?$$

Solution:

$$P(\text{Spam}|\text{Discount}) = \frac{P(\text{Discount}|\text{Spam}) \cdot P(\text{Spam})}{P(\text{Discount})}$$

where:

$$P(\text{Discount}) = P(\text{Discount}|\text{Spam}) \cdot P(\text{Spam}) + P(\text{Discount}|\text{NotSpam}) \cdot P(\text{NotSpam})$$

Substitute the values:

$$P(\text{Discount}) = (0.4 \cdot 0.7) + (0.1 \cdot 0.3) = 0.28 + 0.03 = 0.31$$

$$P(\text{Spam}|\text{Discount}) = \frac{0.4 \cdot 0.7}{0.31} = \frac{0.28}{0.31} \approx 0.903$$

Answer: The probability the email is spam is about 90.3%.

Problem Statement: A factory has two machines. Machine A produces 60% of the products, and Machine B produces 40%. The defect rate for Machine A is 1% ($P(\text{Defect}|A)=0.01$), and for Machine B, it is 5% ($P(\text{Defect}|B)=0.05$). If a defective product is found, what is the probability it was produced by Machine A ($P(A|\text{Defect})$)?

Problem Statement: A factory has two machines. Machine A produces 60% of the products, and Machine B produces 40%. The defect rate for Machine A is 1% ($P(\text{Defect}|A) = 0.01$), and for Machine B, it is 5% ($P(\text{Defect}|B) = 0.05$). If a defective product is found, what is the probability it was produced by Machine A ($P(A|\text{Defect})$)?

Solution:

$$P(A|\text{Defect}) = \frac{P(\text{Defect}|A) \cdot P(A)}{P(\text{Defect})}$$

where:

$$P(\text{Defect}) = P(\text{Defect}|A) \cdot P(A) + P(\text{Defect}|B) \cdot P(B)$$

Substitute the values:

$$P(\text{Defect}) = (0.01 \cdot 0.6) + (0.05 \cdot 0.4) = 0.006 + 0.02 = 0.026$$

$$P(A|\text{Defect}) = \frac{0.01 \cdot 0.6}{0.026} = \frac{0.006}{0.026} \approx 0.231$$

Answer: The probability the defective product was produced by Machine A is about 23.1%.

Two groups are competing for the position on the Board of directors of a corporation. The probabilities that the first and the second group will win are 0.6 and 0.4 respectively. Further, if the first group wins, the probability of introducing a new product is 0.7 and the corresponding probability is 0.3 if the second group wins. Find the probability that the new product introduced was by the second group

Let

E : first group wins

F : second group wins

G : new product is introduced

We need to find the Probability that the new product introduced was by the second group

i.e. **$P(F|G)$**

$$P(F|G) = \frac{P(F).P(G|F)}{P(E).P(G|E) + P(F).P(G|F)}$$

P(E) = Probability that first group wins
= 0.6

P(G|E) = Probability that new product is introduced, if first group wins
= 0.7

P(F) = Probability that the second group wins
= 0.4

P(G|F) = Probability that the a new product is introduced, if second group wins
= 0.3

Putting values in formula,

$$P(F|G) = \frac{0.4 \times 0.3}{0.6 \times 0.7 + 0.4 \times 0.3}$$
$$= \frac{0.12}{0.42 + 0.12}$$

$$= \frac{0.12}{0.54}$$

$$= \frac{2}{9}$$

Therefore, required probability is $\frac{2}{9}$

A weather app predicts rain. Historically, it rains 30% of the time ($P(Rain)=0.3$). The app correctly predicts rain 80% of the time ($P(Prediction=Rain|Rain)=0.8$) but falsely predicts rain 20% of the time ($P(Prediction=Rain|NoRain)=0.2$). If the app predicts rain, what is the probability it will actually rain ($P(Rain|Prediction=Rain)$)?

Problem Statement: A weather app predicts rain. Historically, it rains 30% of the time ($P(Rain) = 0.3$). The app correctly predicts rain 80% of the time ($P(Prediction = Rain|Rain) = 0.8$) but falsely predicts rain 20% of the time ($P(Prediction = Rain|NoRain) = 0.2$). If the app predicts rain, what is the probability it will actually rain ($P(Rain|Prediction = Rain)$)?

Solution:

$$P(Rain|Prediction = Rain) = \frac{P(Prediction = Rain|Rain) \cdot P(Rain)}{P(Prediction = Rain)}$$

where:

$$P(Prediction = Rain) = P(Prediction = Rain|Rain) \cdot P(Rain) + P(Prediction = Rain|NoRain) \cdot P(NoRain)$$

Substitute the values:

$$P(Prediction = Rain) = (0.8 \cdot 0.3) + (0.2 \cdot 0.7) = 0.24 + 0.14 = 0.38$$

$$P(Rain|Prediction = Rain) = \frac{0.8 \cdot 0.3}{0.38} = \frac{0.24}{0.38} \approx 0.632$$

Answer: The probability it will rain is about 63.2%.

Bayes' rule

Following are some applications of Bayes' theorem:

- It is used to calculate the next step of the robot when the already executed step is given.
- Bayes' theorem is helpful in weather forecasting.

NAIVE BAYES CLASSIFIER

- Naïve Bayes algorithm is a **supervised learning** algorithm, which is based on **Bayes theorem** and used for solving classification problems.
- It is a **probabilistic classifier**, which means it **predicts on the basis of the probability of an object**.
- Naïve Bayes Classifier is one of the simple and most effective Classification algorithms which helps in building the fast machine learning models that can make quick predictions.
- Some popular examples of Naïve Bayes Algorithm are **spam filtration, Sentimental analysis, and classifying articles**.

NAIVE BAYES CLASSIFIER

- **Naïve:** It is called Naïve because it assumes that the **occurrence of a certain feature is independent of the occurrence of other features**. Such as if the fruit is identified on the bases of color, shape, and taste, then red, spherical, and sweet fruit is recognized as an apple. Hence each feature individually contributes to identify that it is an apple without depending on each other.
- **Bayes:** It is called Bayes because it depends on the principle of Bayes' Theorem.

NAIVE BAYES CLASSIFIER

■ Suppose we have a dataset of **weather conditions** and corresponding target variable "**Play**". So using this dataset we need to decide that whether we should play or not on a particular day according to the weather conditions. So to solve this problem, we need to follow the below steps:

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

NAIVE BAYES CLASSIFIER

Classification problem may have

k class labels $y_1, y_2, \dots, y_k$ and

n input variables, $X_1, X_2, \dots, X_n$.

We can calculate the conditional probability for a class label with a given instance or set of input values for each column $x_1, x_2, \dots, x_n$ as follows:

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Frame classification as a conditional classification problem with Bayes Theorem as follows:

$$P(y_i | x_1, x_2, \dots, x_n) = P(x_1, x_2, \dots, x_n | y_i) * P(y_i) / P(x_1, x_2, \dots, x_n)$$

NAIVE BAYES CLASSIFIER

Frame classification as a conditional classification problem with Bayes Theorem as follows:

$$P(y_i | x_1, x_2, \dots, x_n) = P(x_1, x_2, \dots, x_n | y_i) * P(y_i) / P(x_1, x_2, \dots, x_n)$$

Consider each input variable as being independent from each other.

This changes the model from a dependent conditional probability model to an independent conditional probability model and dramatically simplifies the calculation.

NAIVE BAYES CLASSIFIER

First, the denominator is removed from the calculation $P(x_1, x_2, \dots, x_n)$ as it is a constant used in calculating the conditional probability of each class for a given instance and has the effect of normalizing the result.

$$P(y_i | x_1, x_2, \dots, x_n) = P(x_1, x_2, \dots, x_n | y_i) * P(y_i)$$

The conditional probability of all variables given the class label is changed into **separate conditional probabilities of each variable value given the class label**.

These independent conditional variables are then multiplied together.

$$P(y_i | x_1, x_2, \dots, x_n) = P(x_1|y_i) * P(x_2|y_i) * \dots P(x_n|y_i) * P(y_i)$$

NAIVE BAYES CLASSIFIER

Working of Naïve Bayes' Classifier:

Working of Naïve Bayes' Classifier can be understood with the help of the below example:

1. Convert the given dataset into frequency tables.
2. Generate Likelihood table by finding the probabilities of given features.
3. Now, use Bayes theorem to calculate the posterior probability.

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

D15	Sunny	Cool	High	Strong	?
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NAIVE BAYES CLASSIFIER - Example

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Calculate Frequency Table

$$P(\text{PlayTennis} = \text{yes}) = 9/14 = .64$$

$$P(\text{PlayTennis} = \text{no}) = 5/14 = .36$$

NAIVE BAYES CLASSIFIER - Example

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Calculate Likelihood Table

$$P(\text{PlayTennis} = \text{yes}) = 9/14 = .64$$

$$P(\text{PlayTennis} = \text{no}) = 5/14 = .36$$

Outlook	Y	N	Humidity	Y	N
sunny	2/9	3/5	high	3/9	4/5
overcast	4/9	0	normal	6/9	1/5
rain	3/9	2/5			
Temperature			Windy		
hot	2/9	2/5	Strong	3/9	3/5
mild	4/9	2/5	Weak	6/9	2/5
cool	3/9	1/5			

NAIVE BAYES CLASSIFIER - Example

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$\langle Outlook = sunny, Temperature = cool, Humidity = high, Wind = strong \rangle$

$$\begin{aligned}
 v_{NB} &= \underset{v_j \in \{yes, no\}}{\operatorname{argmax}} P(v_j) \prod_i P(a_i | v_j) \\
 &= \underset{v_j \in \{yes, no\}}{\operatorname{argmax}} P(v_j) \quad P(Outlook = sunny | v_j) P(Temperature = cool | v_j) \\
 &\quad \cdot P(Humidity = high | v_j) P(Wind = strong | v_j)
 \end{aligned}$$

NAIVE BAYES CLASSIFIER - Example

(Outlook = sunny, Temperature = cool, Humidity = high, Wind = strong)

$$v_{NB} = \underset{v_j \in \{yes, no\}}{\operatorname{argmax}} P(v_j) \prod_i P(a_i | v_j)$$

$$v_{NB}(yes) = P(yes) P(sunny|yes) P(cool|yes) P(high|yes) P(strong|yes)$$

$$= (2/9) * (3/9) * (3/9) * (3/9) * (9/14) = 0.0053$$

Outlook	Y	N		Humidity	Y	N
sunny	2/9	3/5		high	3/9	4/5
overcast	4/9	0		normal	6/9	1/5
rain	3/9	2/5				
Temperature				Windy		
hot	2/9	2/5		Strong	3/9	3/5
mild	4/9	2/5		Weak	6/9	2/5
cool	3/9	1/5				

Probability that we can play the game.

- > $P(\text{Outlook}=\text{Sunny} \mid \text{Play}=\text{Yes}) = 2/9$
- > $P(\text{Temperature}=\text{Cool} \mid \text{Play}=\text{Yes}) = 3/9$
- > $P(\text{Humidity}=\text{High} \mid \text{Play}=\text{Yes}) = 3/9$
- > $P(\text{Wind}=\text{Strong} \mid \text{Play}=\text{Yes}) = 3/9$
- > $P(\text{Play}=\text{Yes}) = 9/14$

NAIVE BAYES CLASSIFIER - Example

(Outlook = sunny, Temperature = cool, Humidity = high, Wind = strong)

$$v_{NB} = \underset{v_j \in \{yes, no\}}{\operatorname{argmax}} P(v_j) \prod_i P(a_i | v_j)$$

$$v_{NB}(no) = P(no) P(sunny|no) P(cool|no) P(high|no) P(strong|no)$$

$$= (3/5) * (1/5) * (4/5) * (3/5) * (5/14) = 0.0206$$

Thus, the naive Bayes classifier assigns the target value **PlayTennis = no to this new instance**, based on the probability estimates learned from the training data.

Outlook	Y	N		Humidity	Y	N
sunny	2/9	3/5		high	3/9	4/5
overcast	4/9	0		normal	6/9	1/5
rain	3/9	2/5				
Temperature				Windy		
hot	2/9	2/5		Strong	3/9	3/5
mild	4/9	2/5		Weak	6/9	2/5
cool	3/9	1/5				

Probability we cannot play a game:

- > $P(\text{Outlook}=\text{Sunny} \mid \text{Play}=\text{No}) = 3/5$
- > $P(\text{Temperature}=\text{Cool} \mid \text{Play}=\text{No}) = 1/5$
- > $P(\text{Humidity}=\text{High} \mid \text{Play}=\text{No}) = 4/5$
- > $P(\text{Wind}=\text{Strong} \mid \text{Play}=\text{No}) = 3/5$
- > $P(\text{Play}=\text{No}) = 5/14$

NAIVE BAYES CLASSIFIER - Example

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Calculate the conditional probability that the target value is **no**, given the **observed attribute** values.

$$v_{NB}(no) = \frac{v_{NB}(no)}{v_{NB}(yes) + v_{NB}(no)} = 0.795$$

$$\frac{.0206}{.0206 + .0053} = .795.$$

NAIVE BAYES CLASSIFIER - Example

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$$v_{NB}(yes) = \frac{v_{NB}(yes)}{v_{NB}(yes) + v_{NB}(no)} = 0.205$$

NAIVE BAYES CLASSIFICATION

The following table gives data set about Diabetic Patient. Using Naïve bayes

Predict for New Data

For a new patient with:

Glucose Level = High

BMI = Overweight

Age = Senior

Record	Glucose Level	BMI	Age	Diabetic
1	High	Overweight	Senior	Yes
2	Normal	Normal	Senior	No
3	Low	Underweight	Adult	No
4	High	Normal	Adult	Yes
5	High	Overweight	Senior	Yes
6	Normal	Normal	Adult	No
7	Low	Underweight	Young	No
8	High	Overweight	Adult	Yes
9	Normal	Normal	Young	No
10	Low	Underweight	Young	No
11	High	Overweight	Senior	Yes
12	Normal	Normal	Adult	No
13	High	Overweight	Senior	Yes
14	Low	Normal	Young	No
15	Normal	Normal	Senior	No

Step 1: Calculate Priors

$$P(\text{Diabetic} = \text{Yes}) = \frac{\text{Count}(\text{Diabetic} = \text{Yes})}{\text{Total Records}} = \frac{6}{15} = 0.4$$

$$P(\text{Diabetic} = \text{No}) = \frac{\text{Count}(\text{Diabetic} = \text{No})}{\text{Total Records}} = \frac{9}{15} = 0.6$$

Step 2: Calculate Likelihoods

For each feature, calculate the likelihood conditioned on being diabetic or not. For example:

For Glucose Level:

- $P(\text{Glucose} = \text{High} \mid \text{Diabetic} = \text{Yes}) = \frac{\text{Count}(\text{High, Yes})}{\text{Count}(\text{Diabetic} = \text{Yes})} = \frac{4}{6} = 0.67$
- $P(\text{Glucose} = \text{High} \mid \text{Diabetic} = \text{No}) = \frac{\text{Count}(\text{High, No})}{\text{Count}(\text{Diabetic} = \text{No})} = \frac{0}{9} = 0.0$

For BMI:

- $P(\text{BMI} = \text{Overweight} \mid \text{Diabetic} = \text{Yes}) = \frac{\text{Count}(\text{Overweight, Yes})}{\text{Count}(\text{Diabetic} = \text{Yes})} = \frac{4}{6} = 0.67$
- $P(\text{BMI} = \text{Overweight} \mid \text{Diabetic} = \text{No}) = \frac{\text{Count}(\text{Overweight, No})}{\text{Count}(\text{Diabetic} = \text{No})} = \frac{1}{9} = 0.11$

For Age:

- $P(\text{Age} = \text{Senior} \mid \text{Diabetic} = \text{Yes}) = \frac{\text{Count}(\text{Senior, Yes})}{\text{Count}(\text{Diabetic} = \text{Yes})} = \frac{3}{6} = 0.5$
- $P(\text{Age} = \text{Senior} \mid \text{Diabetic} = \text{No}) = \frac{\text{Count}(\text{Senior, No})}{\text{Count}(\text{Diabetic} = \text{No})} = \frac{2}{9} = 0.22$

Step 3: Predict for New Data

For a new patient with:

- Glucose Level = High
- BMI = Overweight
- Age = Senior

Calculate posterior probabilities:

1. $P(\text{Diabetic} = \text{Yes} \mid \text{Features})$:

$$P(\text{Diabetic} = \text{Yes} \mid \text{Features}) \propto P(\text{Diabetic} = \text{Yes}) \cdot P(\text{Glucose} = \text{High} \mid \text{Diabetic} = \text{Yes}) \cdot P(\text{BMI} = \text{Overweight} \mid \text{Diabetic} = \text{Yes}) \cdot P(\text{Age} = \text{Senior} \mid \text{Diabetic} = \text{Yes})$$

$$P(\text{Diabetic} = \text{Yes} \mid \text{Features}) \propto 0.4 \cdot 0.67 \cdot 0.67 \cdot 0.5 = 0.0896$$

2. $P(\text{Diabetic} = \text{No} \mid \text{Features})$:

$$P(\text{Diabetic} = \text{No} \mid \text{Features}) \propto P(\text{Diabetic} = \text{No}) \cdot P(\text{Glucose} = \text{High} \mid \text{Diabetic} = \text{No}) \cdot P(\text{BMI} = \text{Overweight} \mid \text{Diabetic} = \text{No}) \cdot P(\text{Age} = \text{Senior} \mid \text{Diabetic} = \text{No})$$

$$P(\text{Diabetic} = \text{No} \mid \text{Features}) \propto 0.6 \cdot 0.0 \cdot 0.11 \cdot 0.22 = 0.0$$

Step 4: Final Prediction

Since $P(\text{Diabetic} = \text{Yes} \mid \text{Features}) > P(\text{Diabetic} = \text{No} \mid \text{Features})$, predict **Diabetic = Yes**.



RA features:

- **Temperature:** Low, Normal, High
- **Vibration Level:** Low, Moderate, High
- **Running Hours:** Short, Medium, Long

Target variable: Maintenance (Yes or No)

Predict if a machine will need maintenance if its Temperature=High, Vibration Level=Low, and Running Hours=Long using Naive Bayes and Bayes' theorem.

Record	Temperature	Vibration Level	Running Hours	Maintenance
1	High	High	Long	Yes
2	Normal	Moderate	Medium	No
3	Low	Low	Short	No
4	High	Moderate	Long	Yes
5	High	High	Long	Yes
6	Normal	Moderate	Medium	No
7	Low	Low	Short	No
8	High	High	Medium	Yes
9	Normal	Low	Medium	No
10	Low	Low	Short	No
11	High	High	Long	Yes
12	Normal	Moderate	Medium	No
13	High	High	Long	Yes
14	Low	Moderate	Medium	No
15	Normal	Moderate	Medium	No

Step 1: Calculate Priors

1. $P(\text{Maintenance} = \text{Yes}) = \frac{\text{Count}(\text{Maintenance} = \text{Yes})}{\text{Total Records}} = \frac{6}{15} = 0.4$
 2. $P(\text{Maintenance} = \text{No}) = \frac{\text{Count}(\text{Maintenance} = \text{No})}{\text{Total Records}} = \frac{9}{15} = 0.6$
-

Step 2: Calculate Likelihoods

For each feature (Temperature, Vibration Level, Running Hours), calculate conditional probabilities based on whether maintenance is **Yes** or **No**.

Temperature:

- $P(\text{Temperature} = \text{High} \mid \text{Maintenance} = \text{Yes}) = \frac{\text{Count}(\text{High, Yes})}{\text{Count}(\text{Maintenance} = \text{Yes})} = \frac{4}{6} = 0.67$
- $P(\text{Temperature} = \text{High} \mid \text{Maintenance} = \text{No}) = \frac{\text{Count}(\text{High, No})}{\text{Count}(\text{Maintenance} = \text{No})} = \frac{0}{9} = 0.0$

Vibration Level:

- $P(\text{Vibration} = \text{High} \mid \text{Maintenance} = \text{Yes}) = \frac{\text{Count}(\text{High, Yes})}{\text{Count}(\text{Maintenance} = \text{Yes})} = \frac{4}{6} = 0.67$
- $P(\text{Vibration} = \text{High} \mid \text{Maintenance} = \text{No}) = \frac{\text{Count}(\text{High, No})}{\text{Count}(\text{Maintenance} = \text{No})} = \frac{0}{9} = 0.0$

Running Hours:

- $P(\text{Running} = \text{Long} \mid \text{Maintenance} = \text{Yes}) = \frac{\text{Count}(\text{Long, Yes})}{\text{Count}(\text{Maintenance} = \text{Yes})} = \frac{3}{6} = 0.5$
- $P(\text{Running} = \text{Long} \mid \text{Maintenance} = \text{No}) = \frac{\text{Count}(\text{Long, No})}{\text{Count}(\text{Maintenance} = \text{No})} = \frac{0}{9} = 0.0$

Step 3: Predict for New Data

For a new machine with:

- Temperature = High
- Vibration Level = High
- Running Hours = Long

Calculate posterior probabilities for both outcomes (Yes and No):

1. $P(\text{Maintenance} = \text{Yes} \mid \text{Features})$:

$$P(\text{Maintenance} = \text{Yes} \mid \text{Features}) \propto P(\text{Maintenance} = \text{Yes}) \cdot P(\text{Temperature} = \text{High} \mid \text{Maintenance} = \text{Yes}) \cdot P(\text{Vibration} = \text{High} \mid \text{Maintenance} = \text{Yes}) \cdot P(\text{Running} = \text{Long} \mid \text{Maintenance} = \text{Yes})$$

$$P(\text{Maintenance} = \text{Yes} \mid \text{Features}) \propto 0.4 \cdot 0.67 \cdot 0.67 \cdot 0.5 = 0.0896$$

2. $P(\text{Maintenance} = \text{No} \mid \text{Features})$:

$$P(\text{Maintenance} = \text{No} \mid \text{Features}) \propto P(\text{Maintenance} = \text{No}) \cdot P(\text{Temperature} = \text{High} \mid \text{Maintenance} = \text{No}) \cdot P(\text{Vibration} = \text{High} \mid \text{Maintenance} = \text{No}) \cdot P(\text{Running} = \text{Long} \mid \text{Maintenance} = \text{No})$$

$$P(\text{Maintenance} = \text{No} \mid \text{Features}) \propto 0.6 \cdot 0.0 \cdot 0.0 \cdot 0.0 = 0.0$$

Step 4: Final Prediction

Since $P(\text{Maintenance} = \text{Yes} \mid \text{Features}) > P(\text{Maintenance} = \text{No} \mid \text{Features})$, the model predicts **Maintenance = Yes**.

NAIVE BAYES CLASSIFIER - Example

The following table gives data set about stolen vehicles. Using Naïve bayes classifier classify the new data (Red, SUV, Domestic).

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

$$p(Yes) = \frac{5}{10} = 0.5$$

$$p(No) = \frac{5}{10} = 0.5$$

Color	Yes	No
Red	3/5	2/5
Yellow	2/5	3/5

Type	Yes	No
Sports	4/5	2/5
SUV	1/5	3/5

Origin	Yes	No
Domestic	2/5	3/5
Imported	3/5	2/5

$$P(Yes|New\ Instance) = p(Yes) * P(Color = Red|Yes) * P(Type = SUV|Yes) * P(Origin = Domestic|Yes)$$

$$P(Yes|New\ Instance) = \frac{5}{10} * \frac{3}{5} * \frac{1}{5} * \frac{2}{5} = \frac{3}{125} = 0.024$$

$$P(No|New\ Instance) = p(No) * P(Color = Red|No) * P(Type = SUV|No) * P(Origin = Domestic|No)$$

$$P(No|New\ Instance) = \frac{5}{10} * \frac{2}{5} * \frac{3}{5} * \frac{3}{5} = \frac{9}{125} = 0.072$$

NAIVE BAYES CLASSIFIER - Example

- b. Estimate conditional probabilities of each attributes {colour, legs, height, smelly} for the species classes: {M, H} using the data given in the table. Using these probabilities estimate the probability values for the new instance – (Colour = Green, Legs = 2, Height = Tall and Smelly = No) (10 Marks)

No	Colour	Legs	Height	Smelly	Species
1	White	3	Short	Yes	M
2	Green	2	Tall	No	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	No	H
6	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

Text Classification

Naïve Bayes Model – Text Classification Example

- Dataset for the text classification with training and test data is given below.
- The goal is to classify the test data into the right class as h or —h (read as not h).

	Document ID	Keywords in the document	Class h
Training Set	1	Love Happy Joy Joy Happy	Yes
	2	Happy Love Kick Joy Happy	Yes
	3	Love Move Joy Good	Yes
	4	Love Happy Joy Love Pain	Yes
	5	Joy Love Pain Kick Pain	No
	6	Pain Pain Love kick	No
Testing Set	7	Love Pain Joy Love Kick	?

1. Understand the Dataset

- **Training Set:** Contains documents with keywords and their corresponding class labels.
- **Testing Set:** Contains documents without a class label (to be predicted).

2. Steps to Use Naive Bayes

- **Feature Extraction:** Convert the text data into numerical format (e.g., frequency of keywords)
- **Prior Probabilities:** Calculate the prior probability for each class (e.g., $P(Yes)$ and $P(No)$).
- **Likelihood:** Compute the likelihood $P(Word|Class)$ for each keyword.
- **Prediction:** Use the Naive Bayes formula to calculate the posterior probability for the test document belonging to each class and assign the class with the highest probability.

$$P(Class|Document) \propto P(Class) \times \prod P(Word|Class)$$

3. Calculation

Let's assume:

- **Training Keywords:** "Love", "Happy", "Joy", "Pain", "Kick", "Move", "Good".
- **Classes:** "Yes", "No".

Step 1: Count Frequencies

For each keyword, count how often it appears in the documents for each class.

Step 2: Calculate Probabilities

Using Laplace Smoothing:

$$P(Word|Class) = \frac{\text{Frequency of Word in Class} + 1}{\text{Total Words in Class} + \text{Vocabulary Size}}$$

Step 3: Apply Naive Bayes Formula

Multiply the probabilities for all the words in Document 7 for each class and normalize.

1. Training Data Overview

We have six training documents (IDs 1-6) and two classes: "Yes" and "No".

Class counts:

- **Yes:** Documents 1, 2, 3, 4 → 4 documents.
- **No:** Documents 5, 6 → 2 documents.

Prior probabilities:

$$P(Yes) = \frac{4}{6} = 0.667, \quad P(No) = \frac{2}{6} = 0.333$$

2. Keyword Frequencies

Extract the vocabulary and count the occurrences of each word in the training data for each class.

Keyword	Yes Count	No Count
Love	5	2
Happy	3	0
Joy	4	1
Pain	1	3
Kick	1	2
Move	1	0
Good	1	0

3. Calculate Likelihoods

Using Laplace smoothing:

$$P(Word|Class) = \frac{\text{Frequency of Word in Class} + 1}{\text{Total Words in Class} + \text{Vocabulary Size}}$$

- Vocabulary size = 7 (unique keywords).
- Total words in "Yes" documents = 15.
- Total words in "No" documents = 8.



Likelihoods for "Yes":

$$P(\text{Love}|\text{Yes}) = \frac{5+1}{15+7} = \frac{6}{22} = 0.273$$

$$P(\text{Happy}|\text{Yes}) = \frac{3+1}{15+7} = \frac{4}{22} = 0.182$$

$$P(\text{Joy}|\text{Yes}) = \frac{4+1}{15+7} = \frac{5}{22} = 0.227$$

$$P(\text{Pain}|\text{Yes}) = \frac{1+1}{15+7} = \frac{2}{22} = 0.091$$

$$P(\text{Kick}|\text{Yes}) = \frac{1+1}{15+7} = \frac{2}{22} = 0.091$$

Likelihoods for "No":

$$P(\text{Love}|\text{No}) = \frac{2+1}{8+7} = \frac{3}{15} = 0.2$$

$$P(\text{Happy}|\text{No}) = \frac{0+1}{8+7} = \frac{1}{15} = 0.067$$

$$P(\text{Joy}|\text{No}) = \frac{1+1}{8+7} = \frac{2}{15} = 0.133$$

$$P(\text{Pain}|\text{No}) = \frac{3+1}{8+7} = \frac{4}{15} = 0.267$$

$$P(\text{Kick}|\text{No}) = \frac{2+1}{8+7} = \frac{3}{15} = 0.2$$

4. Apply Naive Bayes Formula

Document 7: "Love Pain Joy Love Kick"

We calculate probabilities for both classes by multiplying prior probabilities and likelihoods of the words in the document.

For "Yes":

$$P(Yes|Doc7) \propto P(Yes) \times P(Love|Yes)^2 \times P(Pain|Yes) \times P(Joy|Yes) \times P(Kick|Yes)$$

$$P(Yes|Doc7) \propto 0.667 \times (0.273)^2 \times 0.091 \times 0.227 \times 0.091$$

$$P(Yes|Doc7) \propto 0.667 \times 0.074 \times 0.091 \times 0.227 \times 0.091 = 0.000102$$

For "No":

$$P(No|Doc7) \propto P(No) \times P(Love|No)^2 \times P(Pain|No) \times P(Joy|No) \times P(Kick|No)$$

$$P(No|Doc7) \propto 0.333 \times (0.2)^2 \times 0.267 \times 0.133 \times 0.2$$

$$P(No|Doc7) \propto 0.333 \times 0.04 \times 0.267 \times 0.133 \times 0.2 = 0.000095$$

5. Normalize Probabilities

$$P(Yes|Doc7) = \frac{0.000102}{0.000102 + 0.000095} = 0.518$$

$$P(No|Doc7) = \frac{0.000095}{0.000102 + 0.000095} = 0.482$$

6. Prediction

Since $P(Yes|Doc7) > P(No|Doc7)$, the predicted class for Document 7 is "Yes".

Example 2

Dataset

Training Set:

Document ID	Text (Keywords)	Class
1	good excellent awesome happy	Positive
2	bad terrible awful hate	Negative
3	excellent happy good perfect	Positive
4	terrible hate awful bad	Negative
5	awesome perfect great good	Positive

Testing Set:

Document ID	Text (Keywords)	Class
6	good perfect terrible bad	?

Dataset

Training Set:

Document ID	Text (Keywords)	Class
1	good excellent awesome happy	Positive
2	bad terrible awful hate	Negative
3	excellent happy good perfect	Positive
4	terrible hate awful bad	Negative
5	awesome perfect great good	Positive

Testing Set:

Document ID	Text (Keywords)	Class
6	good perfect terrible bad	?

Steps:

1. Prior Probabilities

We count how many documents belong to each class.

- **Positive:** 3 documents (1, 3, 5).
- **Negative:** 2 documents (2, 4).

Prior probabilities:

$$P(\text{Positive}) = \frac{3}{5} = 0.6, \quad P(\text{Negative}) = \frac{2}{5} = 0.4$$

Dataset

Training Set:

Document ID	Text (Keywords)	Class
1	good excellent awesome happy	Positive
2	bad terrible awful hate	Negative
3	excellent happy good perfect	Positive
4	terrible hate awful bad	Negative
5	awesome perfect great good	Positive

Testing Set:

Document ID	Text (Keywords)	Class
6	good perfect terrible bad	?

Serial No.	Keyword	Positive Count	Negative Count
1	good	3	0
2	excellent	2	0
3	awesome	2	0
4	happy	2	0
5	perfect	2	0
6	bad	0	2
7	terrible	0	2
8	awful	0	2
9	hate	0	2
10	great	1	0

Testing Set:

Document ID	Text (Keywords)	Class
6	good perfect terrible bad	?

Serial No.	Keyword	Positive Count	Negative Count
1	good	3	0
2	excellent	2	0
3	awesome	2	0
4	happy	2	0
5	perfect	2	0
6	bad	0	2
7	terrible	0	2
8	awful	0	2
9	hate	0	2
10	great	1	0

3. Likelihoods with Laplace Smoothing

Using Laplace smoothing:

$$P(\text{Word}|\text{Class}) = \frac{\text{Frequency of Word in Class} + 1}{\text{Total Words in Class} + \text{Vocabulary Size}}$$

- Vocabulary size = 10 (unique keywords).
- Total words in "Positive" documents = 12.
- Total words in "Negative" documents = 8.

Likelihoods for "Positive":

$$P(\text{good}|\text{Positive}) = \frac{3 + 1}{12 + 10} = \frac{4}{22} = 0.182$$

$$P(\text{excellent}|\text{Positive}) = \frac{2 + 1}{12 + 10} = \frac{3}{22} = 0.136$$

$$P(\text{bad}|\text{Positive}) = \frac{0 + 1}{12 + 10} = \frac{1}{22} = 0.045$$

(Similarly for all other keywords.)

Likelihoods for "Negative":

$$P(\text{good}|\text{Negative}) = \frac{0 + 1}{8 + 10} = \frac{1}{18} = 0.056$$

$$P(\text{bad}|\text{Negative}) = \frac{2 + 1}{8 + 10} = \frac{3}{18} = 0.167$$

$$P(\text{perfect}|\text{Negative}) = \frac{0 + 1}{8 + 10} = \frac{1}{18} = 0.056$$

Serial No.	Keyword	Positive Count	Negative Count
1	good	3	0
2	excellent	2	0
3	awesome	2	0
4	happy	2	0
5	perfect	2	0
6	bad	0	2
7	terrible	0	2
8	awful	0	2
9	hate	0	2
10	great	1	0

- For **Positive** class:

$$P(\text{terrible} \mid \text{Positive}) = \frac{0 + 1}{12 + 10} = \frac{1}{22} \approx 0.0455$$

- For **Negative** class:

$$P(\text{terrible} \mid \text{Negative}) = \frac{2 + 1}{8 + 10} = \frac{3}{18} \approx 0.1667$$

Results:

- $P(\text{terrible} \mid \text{Positive}) = 0.0455$
- $P(\text{terrible} \mid \text{Negative}) = 0.1667$



Document ID	Text (Keywords)	Class
6	good perfect terrible bad	?

4. Classify Document 6: "good perfect terrible bad"

For Positive:

$$P(\text{Positive}|\text{Doc6}) \propto P(\text{Positive}) \times P(\text{good}|\text{Positive}) \times P(\text{perfect}|\text{Positive}) \times P(\text{terrible}|\text{Positive}) \times P(\text{bad}|\text{Positive})$$

$$P(\text{Positive}|\text{Doc6}) \propto 0.6 \times 0.182 \times 0.136 \times 0.045 \times 0.045 = 0.000300$$

For Negative:

$$P(\text{Negative}|\text{Doc6}) \propto P(\text{Negative}) \times P(\text{good}|\text{Negative}) \times P(\text{perfect}|\text{Negative}) \times P(\text{terrible}|\text{Negative}) \times P(\text{bad}|\text{Negative})$$

$$P(\text{Negative}|\text{Doc6}) \propto 0.4 \times 0.056 \times 0.056 \times 0.167 \times 0.167 = 0.000087$$

Since $P(\text{Positive}|\text{Doc6}) > P(\text{Negative}|\text{Doc6})$, the predicted class for Document 6 is "Positive".